# Graphs

## Count. Show it in a graph. Ask questions.

### Tally marks: count by 5s

| | | |
|---|---|---|
| apples | 𝍸 \|\| | 7 |
| cookies | \|\|\|\| | 4 |
| cupcakes | 𝍸 | 5 |

4 lines and a slash is 5.
Count 5, then count on: 5, 6, 7.

### Picture graph

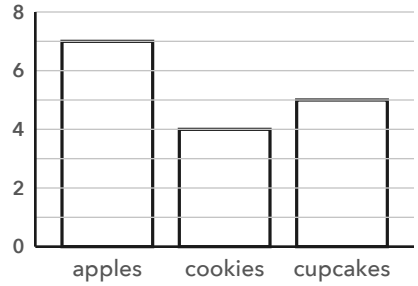
**Our favorite snack**

| | |
|---|---|
| apples | ☆☆☆☆☆☆☆ |
| cookies | ☆☆☆☆ |
| cupcakes | ☆☆☆☆☆ |

Each ☆ = 1

### Bar graph

**Our favorite snack**




### Ask questions

$7 - 4 = 3$ 3 more apples than cookies

$7 + 4 + 5 = 16$ 16 kids in all

**Your turn!**

How many more cupcakes than cookies? **1**

How many like cupcakes or cookies? **9**

**For teachers**

Goal: read and make tally charts, picture graphs and bar graphs, and answer how many, how many more and how many in all (2.MD.10). X charts show lengths (2.MD.9). Make graphs of the group's real data.

# Graphs: Practice

Read the graph carefully. Count the rows or the squares.

**Part 1** Count the tally marks

1)  7

2)  13

3)  4

4)  18

5)  11

6)  9

7)  17

8)  6

**Part 2** Draw tally marks Make groups of 5.

9) 8

10) 14

11) 3

12) 12

13) 19

14) 10

## Graphs: Practice (continued)

### Part 3 Picture graphs Each picture is 1.

15–18)

**Pets at home**




How many kids have dogs? **6**

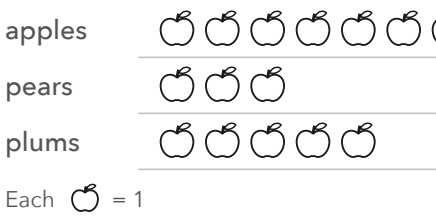
How many more fish than cats? **3**

Dogs and fish together? **13**

How many in all? **17**

19–22)

**Fruit we ate**




How many kids ate plums? **5**

How many more apples than pears? **5**

Pears and apples together? **11**

How many in all? **16**

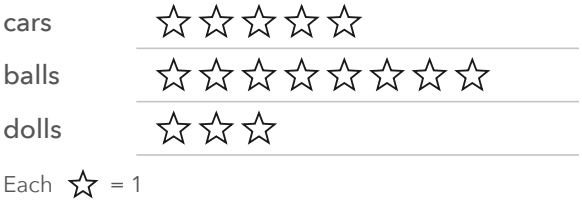
### Part 4 Make a picture graph Draw a circle for each tally mark.

23)

**Toys in the box**




**Make the picture graph**

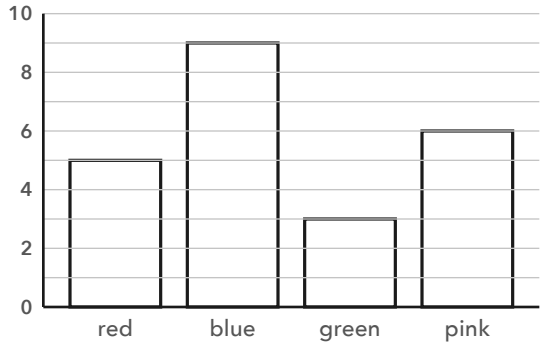


## Graphs: Practice (continued)

**Part 5** Bar graphs Read across from the top of the bar.

24–27)

**Favorite color**




How many kids like red? **5**

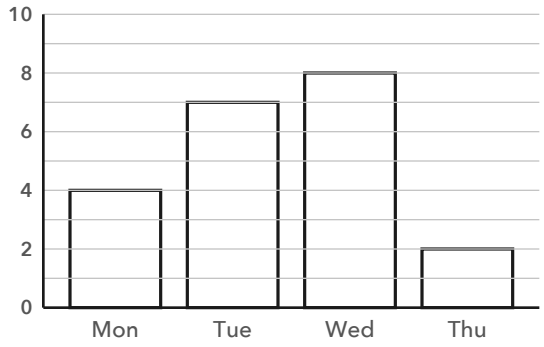
How many more blue than green? **6**

Pink and blue together? **15**

How many in all? **23**

28–31)

**Books we read**




How many books on Tue? **7**

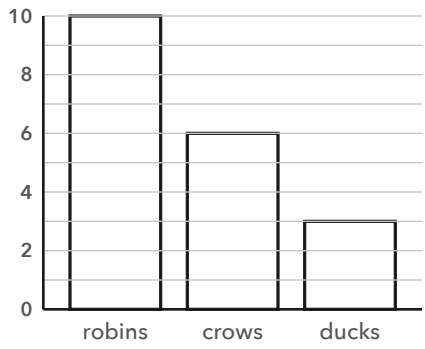
How many more Wed than Thu? **6**

Thu and Mon together? **6**

How many in all? **21**

32–35)

**Birds we saw**




How many were crows? **6**

How many more robins than ducks? **7**

Crows and ducks together? **9**

How many in all? **19**

## Graphs: Practice (continued)

**Part 6** Make a bar graph Color one square for each tally mark.

36)

**Snacks sold**

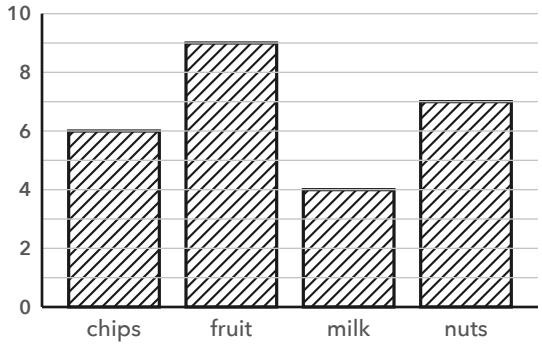
chips 𝍸 |

fruit 𝍸 ||||

milk ||||

nuts 𝍸 ||

**Make the bar graph**




**Part 7** X charts Each X is one thing that was measured.

37–39)

**Pencil lengths (inches)**

| 3 in | 4 in | 5 in | 6 in | 7 in |
|---|---|---|---|---|
| X | X | X | | |
| X | X | X | | |
| | X | X | | |
| | X | | | |
| | | X | X | X |

How many pencils are 4 in long? **5**

How many are longer than 5 in? **3**

How many pencils in all? **13**

40–42)

**Leaf lengths (cm)**

| 5 cm | 6 cm | 7 cm | 8 cm | 9 cm |
|---|---|---|---|---|
| X | X | X | X | X |
| | X | X | X | |
| | X | X | X | |
| | | X | X | |
| | | | | X |

How many leaves are 7 cm long? **4**

How many are shorter than 7 cm? **4**

How many leaves in all? **14**

## Graphs: Practice (continued)

**Part 8** Count the tally marks

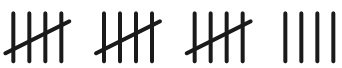
43)  3

44)  5

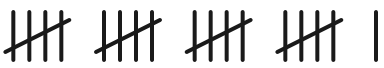
45)  8

46)  12

47)  16

48)  19

49)  20

50)  21

51)  22

52)  23

53)  24

54)  25

**Part 9** Draw tally marks

55) 4

56) 5 

## Graphs: Practice (continued)

**Part 9** Draw tally marks (continued)

57) 6 

58) 7 

59) 9 

60) 15 

61) 16 

62) 18 

63) 20 

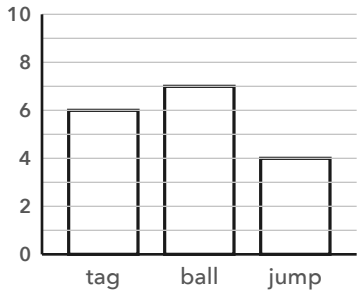
64) 21 

65) 22 

66) 23 

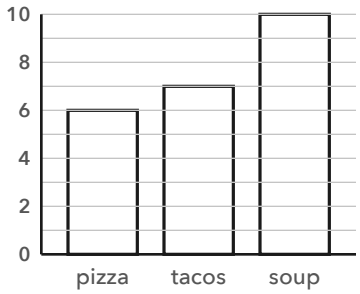
**Part 10** Read the bar graph

67) **Recess games**




How many more ball than jump? 3

68) **Lunch**

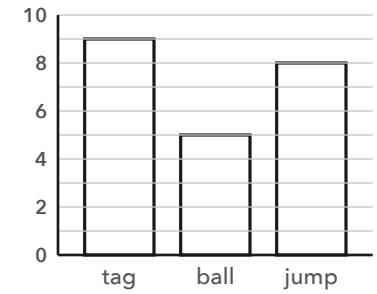



Tacos and soup together? 17

## Graphs: Practice (continued)

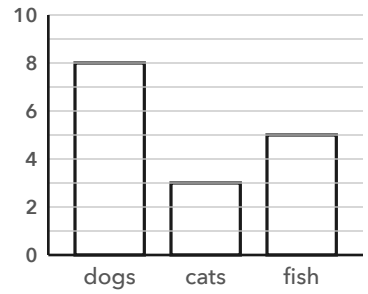
### Part 10 Read the bar graph (continued)

69)




How many more tag than ball? 4

70)




How many kids picked fish? 5

### Part 11 Remember? Problems from earlier packets.

71) 30 cm + 25 cm = 55 cm

72) half past 4 = 4:30

73) 3 quarters = 75¢

74) 800 − 356 = 444

75) 4 fourths = 1 whole

76) half of 78 = 39

### Part 12 Word problems Make a tally chart if it helps.

77) A bar graph shows 9 kids like soccer and 6 kids like tag. How many more kids like soccer?

Number sentence 6 + 3 = 9

Answer 3 kids

Draw or show your work

78) A tally chart shows 3 groups of 5 and 2 more marks. How many marks is that?

Number sentence 5 + 5 + 5 + 2 = 17

Answer 17 marks

Draw or show your work

## Graphs: Practice (continued)

### Part 12 Word problems (continued)

79) A picture graph shows 7 cats and 8 dogs. How many pets in all?

Number sentence $7 + 8 = 15$

Answer 15 pets

Draw or show your work

80) A class voted: 12 for pizza, 5 for tacos. How many fewer votes did tacos get?

Number sentence $5 + 7 = 12$

Answer 7 votes

Draw or show your work

81) Challenge: 20 kids voted: 8 for red, 5 for blue, the rest for green. How many voted for green?

Number sentence $8 + 5 = 13, 13 + 7 = 20$

Answer 7 kids

Draw or show your work

# Check-In: Graphs

Show what you know. No help this time.

**Mastery: 7 of 8 (80%)**

**Part 1** Count the tally marks

1) 𝍸 𝍸 **10**

2) 𝍸 𝍸 𝍸 **15**

3) 𝍸 𝍸 |||| **14**

**Part 2** Draw tally marks

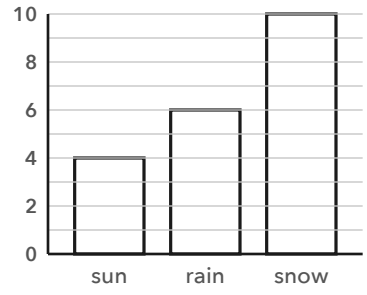
4) **13** 𝍸 𝍸 |||

5) **17** 𝍸 𝍸 𝍸 ||

6) **11** 𝍸 𝍸 |

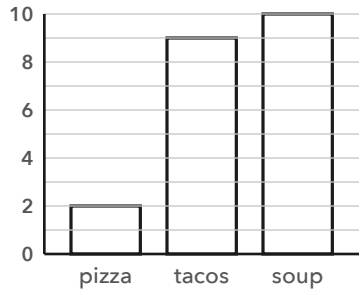
**Part 3** Read the bar graph

7) **Weather**




How many in all? **20**

8) **Lunch**




Soup and pizza together? **12**